6.5 EMF EQUATION

We have known that EMFs are induced in the transformers primary and secondary windings when alternating flux link both of them. The frequency of alternation of the flux will depend upon the frequency of the primary supply voltage which is normally 50 Hz. The magnitude of the EMF induced in the two windings of a transformer will be different if they have different number of turns. Let us now derive the EMF equation. Referring to Fig. 6.6, and considering a sinusoidal input voltage, V_1 at a frequency, f, the flux produced due to current, I_0 is $\phi = \phi_m \sin \omega t$.

The general equation for the instantantaneous value of the EMF induced, e is expressed as

$$e = -N\frac{d\phi}{dt}$$

[Note that the minus sign indicates that the induced EMF opposes the supply voltage according to Lenz's law.]

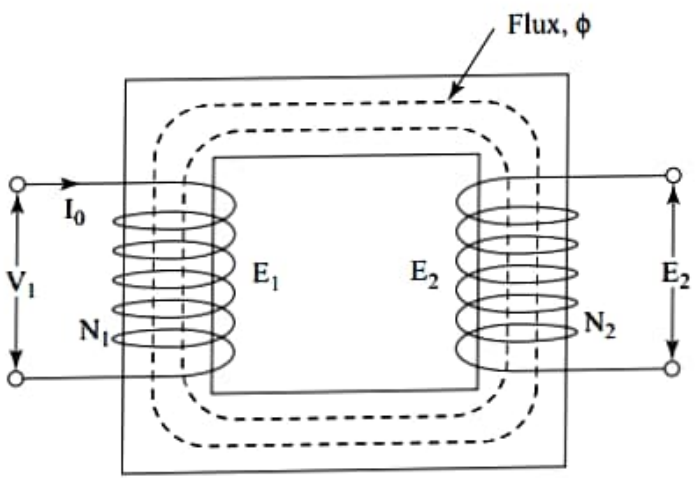


Figure 6.6 *Two windings of a transformer wound on a common core*

A sinusoidally varying flux is represented as

$$\phi = \phi_m \sin \omega t$$

where ϕ is the instantaneous value, and ϕ_m is the maximum value

Considering $\phi = \phi_m \sin \omega t$,

$$e = -N \frac{d(\phi_m \sin \omega t)}{dt}$$

we get

$$\begin{aligned} e &= -N \phi_m \omega \cos \omega t \\ &= -N \phi_m \omega [-\sin (\omega t - \pi/2)] \\ &= N \phi_m \omega \sin (\omega t - 90°) \end{aligned}$$

This equation is of the form

$$e = E_m \sin (\omega t - 90°)$$

where $E_m = N \phi_m \omega$

or, $E_m = N \phi_m 2\pi f$

We know for a sinusoidal voltage, the RMS value is $\frac{1}{\sqrt{2}}$ times its maximum value. If, E is the RMS value of the induced EMF, then

$$E = \frac{E_m}{\sqrt{2}} = \frac{N\phi_m 2\pi f}{\sqrt{2}} = \frac{N\phi_m 2 \times 3.14 f}{1.414}$$

or, $E = 4.44 \phi_m f N V$ (6.2)

Primary winding has N_1 turns. The induced EMF in the primary winding E_1 is

$$E_1 = 4.44\, \phi_m\, f\, N_1\, V \tag{6.3}$$

Since the same flux producing E_1 also links the secondary winding having N_2 number of turns, the induced EMF in the secondary winding, E_2 is

$$E_2 = 4.44\, \phi_m\, f\, N_2\, V \tag{6.4}$$

Dividing eq. (6.3) by eq. (6.4),

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

or,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

K is called the ratio of voltage transformation. For a step-up transformer, $N_2 > N_1$, and hence, the value of K is more than 1. For a step-down transformer, N_2 is less than N_1, and hence $K < 1$.

Example 6.4 A 100 kVA, 1100/220 V, 50 Hz transformer has 100 turns on its secondary winding. Calculate the number of turns of the primary winding; the currents that would flow in both the windings when fully loaded, and the maximum value of flux in the core.

Solution:

Given, $N_2 = 100$

$V_1 \approx E_1$ and $V_2 \approx E_2$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$N_1 = N_2\left(\frac{E_1}{E_2}\right) = 100\left(\frac{1100}{220}\right) = 500 \text{ turns}$$

Rating of the transformer is = 100 kVA = 100×10^3 VA.

Primary current, $$I_1 = \frac{VA}{V_1} = \frac{100 \times 10^3}{1100} = 90.1 \text{ A}$$

Since volt-ampere rating is the same for the transformer on both the sides,

scondary current, $$I_2 = \frac{VA}{V_2} = \frac{100 \times 10^3}{220} = 450.5 \text{ A}$$

For calculating ϕ_m, we will use the EMF equation.

$$E_1 = 4.44\, \phi_m\, f\, N_1$$

Substituting values $$1100 = 4.44\, \phi_m \times 50 \times 500$$

or, $$\phi_m = 9.9 \times 10^{-3} \text{ Wb}$$

Example 6.2 A transformer has 900 turns on its primary winding and 300 turns on its secondary. A voltage of 230 V at 50 Hz is connected across its primary winding. The cross-sectional area of the core is 64 cm^2. Calculate the magnitude of the induced EMF in the secondary winding. Also calculate the value of maximum flux density in the core.

Solution:

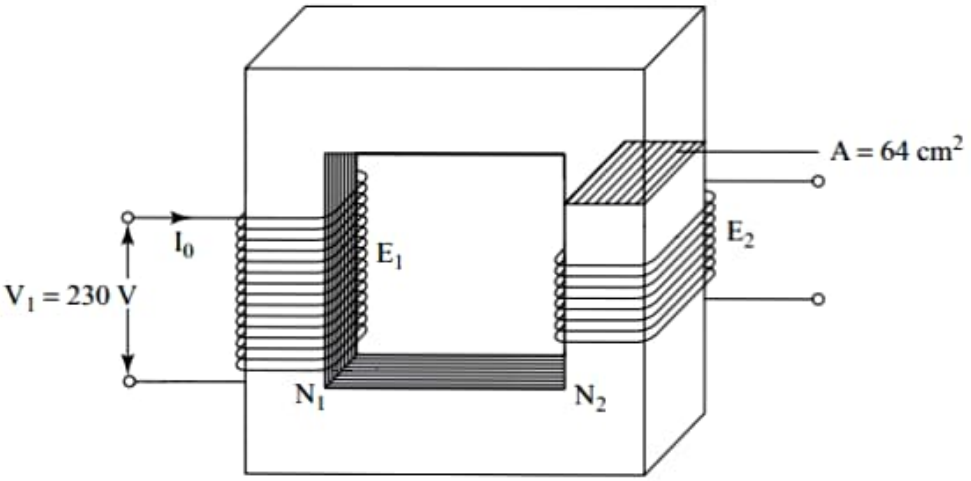


Figure 6.22

Given $V_1 = 230$ V.

The induced EMF in the primary windings is E_1. E_1 is slightly less than V_1 because there will be some voltage drop in the winding.

$V_1 > E_1$, $V_1 - E_1$ = Voltage drop in the primary winding due to current, I_0 flowing through it.

The no-load current, I_0 is very small as compared to the current that would flow when some electrical load is connected across the secondary winding. Here, the transformer is on no-load, i.e., no load has been connected to its secondary winding.

If we neglect the no-load voltage drop in the winding, we can write $V_1 = E_1$.

$$E_1 = 4.44\,\phi_m\,f\,N_1$$

and

$$E_2 = 4.44\,\phi_m\,f\,N_2$$

therefore,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

or,

$$E_2 = E_1\left(\frac{N_2}{N_1}\right) = 230\left(\frac{300}{900}\right) = 76.7\text{ V}$$

Again,

$$E_1 = 4.44\,\phi_m\,f\,N_1$$

$$= 4.44\,B_m\,A\,f\,N_1$$

when B_m is the maximum flux density and A is the cross-sectional area of the core.

Substituting values

$$230 = 4.44\,B_m \times \frac{64}{10^4} \times 50 \times 900$$

or,

$$B_m = \frac{230 \times 10^4}{4.44 \times 64 \times 45000} = \frac{2300}{4.44 \times 64 \times 45}$$

or,

$$B_m = 0.18\text{ Wb/m}^2$$

Example 7.4 A four-pole 220 V dc shunt generator supplies a load of 3 kW at 220 V. The resistance of the armature winding is 0.1 Ω and that of the field winding is 110 Ω. Calculate the total armature current, the current flowing through armature conductors, and the EMF induced. Assume that the armature winding is wave wound.

Solution:

Output power = 3 kW = 3000 W

Power = output voltage, V × Output current, I_L

$$I_L = \frac{3000}{220} = 13.6\ \text{A}$$

From the figure it can be seen that

$$I_f = \frac{V}{R_f} = \frac{220}{110} = 2\ \text{A}$$

$$I_a = I_L + I_f$$

$$= 13.6 + 2 = 15.6\ \text{A}$$

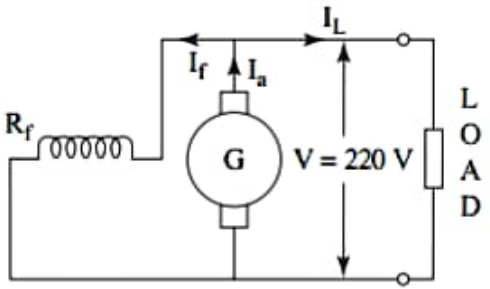


Figure 7.28

The armature winding is wave wound. The number of parallel paths is 2. That is, all the armature conductors are connected in such a way that half the armature current flows through each path. Thus, current flowing though each armature conductor will be $I_a/2$ i.e., $\frac{15.6}{2} = 7.8$ Amps. E is the EMF induced in the armature. A voltage drop of I_aR_a takes place in the armature winding when it is supplying current. The remaining voltage, V is available across the load terminals.
Thus,

$$E - I_aR_a = V$$

or,

$$E = V + I_aR_a = 220 + 15.6 \times 0.1 = 218.44 \text{ V}$$